

ZEPTO DATA ANALYSIS PROJECT

SQL exploration, cleaning & Power BI dashboard

Executive Summary



Scope

Analyze product dataset for categories, pricing, discounts, stock, inventory, estimated revenue



Approach

SQL cleaning + Power BI visualizations



Key outputs

High-discount lists, high-MRP OOS, category revenue, price/gram, inventory weight

Dataset Overview



Rows / Columns

3,732 rows · 9 columns



Core fields

category, name, mrp, discountPercent, discountedSellingPrice



Primary key

sku_id



Inventory fields

weightInGms, availableQuantity, outOfStock, quantity

Project Objectives

Quality & Structure

Understand dataset shape, NULLs, duplicates, zero prices

Cleaning

Remove zero-MRP rows, convert paise
→ rupees

Analysis

Discounts, pricing, price/gram, inventory weight, estimated revenue

Dashboard

Power BI summaries for categories, pricing, stock, KPIs

Methodology

Pipeline

Load → Explore → Clean → Analyze → Visualize

1. Data loading into PostgreSQL
2. Exploration: counts, samples, distinct categories, stock status
3. Quality checks: NULLs, zero prices, duplicate names
4. Cleaning: DELETE zero MRP; convert paise to rupees (mrp/100)
5. SQL analysis and Power BI dashboarding

SQL Analyses Performed

01

Top discounted products

Identify largest discountPercent

03

Estimated revenue

discountedSellingPrice × availableQuantity by category

02

High-MRP OOS

Out-of-stock items with MRP > ₹300

04

Price per gram & weight classes

discountedSellingPrice / weightInGms; Low/Medium/Bulk groups

Key Insights

Discounts

Discount percent surfaces strongest promotions

Availability risk

High-MRP OOS items highlight availability gaps

Normalized value

Price/gram enables fair cross-weight comparison

Inventory mass

Total inventory weight reveals physical stock distribution

Power BI Dashboard

Dashboard components

- Category-level analysis
- Pricing & discount visuals
- Stock & availability panels
- Inventory metrics and KPIs
- Interactive filters (category, stock, price, discount)



Source

PBIX file (supplied)



Interactivity

Slicers for category, stock, price range, discount range

Technical Implementation



Database

PostgreSQL / SQL



Dataset

CSV → zepto table



Core SQL

SELECT, WHERE, GROUP BY, HAVING, ORDER BY, CASE,
aggregates



Cleaning ops

DELETE zero MRP; UPDATE mrp = mrp/100;
discountedSellingPrice = /100

Conclusion & Future Scope

1

Conclusion

End-to-end example: SQL cleans + queries → Power BI communicates insights

2

Future scope

Add time-series sales, customer/order data, automated refresh pipelines

3

Next steps

Enable trend analysis, stock-change tracking, actual revenue from orders